

Inter-Avatar Distance and Conversation Satisfaction in Immersive Virtual Reality

: Exploring the Moderating Effects of the Sense of Embodiment

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2023. 05. 12.

Index

1. Introduction
2. Literature review
3. Methods
4. Results
5. Discussions

1. Introduction

- In recent years, virtual reality (VR) technology has emerged as a novel way for individuals to engage in social interaction within virtual environments that simulate real-world experiences.
- As this technology continues to advance, it has become increasingly important to understand how people interact with each other in these virtual spaces.
- Avatars, which are digital representations of users within the VR, are crucial for facilitating interaction between people.
- However, despite a substantial body of literature on the effect of interpersonal distance on communication and social interaction in real-world settings, few studies have investigated the impact of avatar distance on communication in immersive virtual environments (IVE).
- The main concern of this study is to investigate whether inter-avatar distance has a similar effect on communication in IVE, just as interpersonal distance affects conversation satisfaction in the physical environment.

2. Literature review

(1) Space and interpersonal communication

- The concept of personal space has been defined since the 1960s and refers to an invisible area around a person (Felipe & Sommer, 1966; Sommer, 1969).
- It is crucial to respect personal space during interpersonal interactions. However, while the invasion of personal space can cause tension and anxiety (Felipe & Sommer, 1966), it is uncertain whether violating personal space brings positive or negative effects to people.

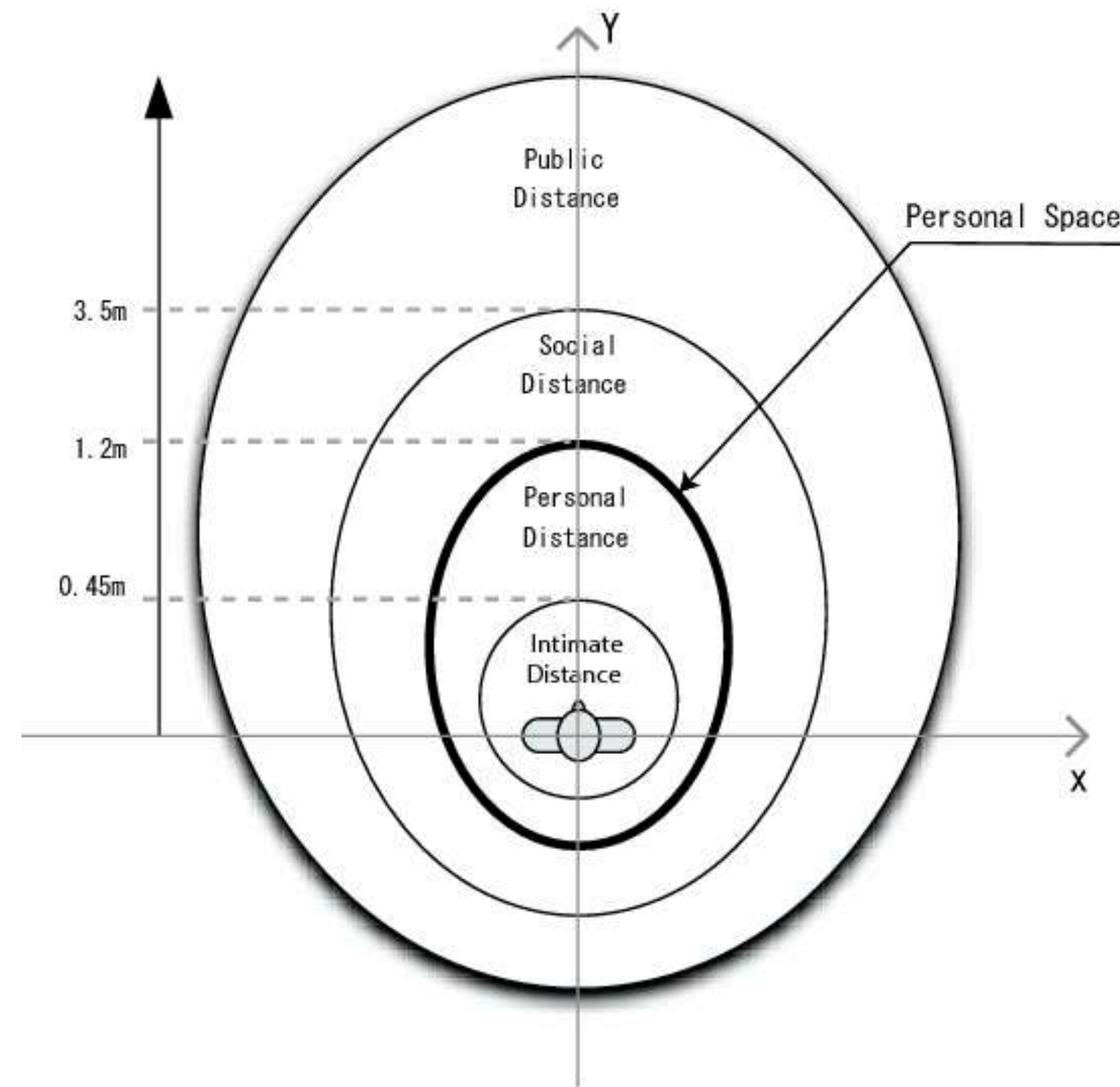


Fig. 1 The intimate distance (0.3 m)



Fig. 2 The personal distance (1.20 m)

▲ image from https://doi.org/10.1007/978-3-642-04052-8_6

▲ image from <https://doi.org/10.1007/s10055-010-0169-3>

2. Literature review

(1) Space and interpersonal communication

- According to nonverbal expectancy violation theory (Burgoon, 1978), people make assumptions about others' nonverbal cues, including the distance they maintain during communication. If the partner violates the expectation, it may cause unintended consequences.
- Several studies have confirmed the existence of personal space in VEs (e.g., Bailenson et al., 2001; Chad et al., 2006; Nassiri et al., 2005; Nassiri et al., 2010), with users typically maintaining a distance similar to that in the real world.
- Such prior works suggest that distance perception similar to the real world can be fully experienced, even in the IVE. We define "inter-avatar distance" and propose the following hypothesis:

➡ **H1: Conversation satisfaction will differ according to the distance between users in the IVE.**

2. Literature review

(2) Media factors influencing communication outcomes: **embodiment**

- In VR communication, the characteristics of VR as a media will certainly have an impact on the communication, so we would like to first explore media factors.
 - The sense of embodiment was defined as an object "E is embodied if and only if some of E's attributes are changed in the same way as the properties of one's body." (De Vignemont, 2011).
 - Intense feelings of self-presence during virtual experiences may cause confusion about identity or reality (Lee, 2004).
 - Several studies have found that embodiment can help resolve cognitive dissonance in the virtual space, which suggests a relationship between embodiment and sense of space (Slater and Usoh, 1994).
 - Thus, a higher sense of embodiment can be inferred to make it possible for users to feel more satisfied.
- ➡ **H2: Conversation satisfaction will differ according to the user's sense of embodiment in the IVE.**
- ➡ **H3: The effect of distance on conversation satisfaction will vary according to the user's sense of embodiment.**

2. Literature review

(3) Individual factors influencing communication outcomes: **user's gender**

- Meanwhile, demographic differences will inevitably affect the quality of VR communication.
 - Nassiri et al. (2005) found that anxiety levels related to personal space invasion were affected by the gender combination of both the invading avatar and the avatar being invaded. These effects were more significant in the virtual world compared to the actual world.
 - Bailenson et al. (2001) investigated mutual gaze and personal space in VE, finding that female participants maintained a greater distance from virtual agents with whom they made eye contact, suggesting that they are less vigilant about personal space than males.
 - In short, it can be surmised that there is also some interaction effect among some personal space along with gender.
- ➡ **H4: Conversation satisfaction will differ according to the user's gender in the IVE.**
- ➡ **H5: The effect of distance on conversation satisfaction will vary according to the user's gender.**

3. Methods

(1) Participants

- The participants were randomly assigned and each participant was given extra credit for a course as an incentive for participation and was provided informed consent.
- Among the 61 participants, 60.7% were females, and aged 17-25 years ($M = 20.3$).

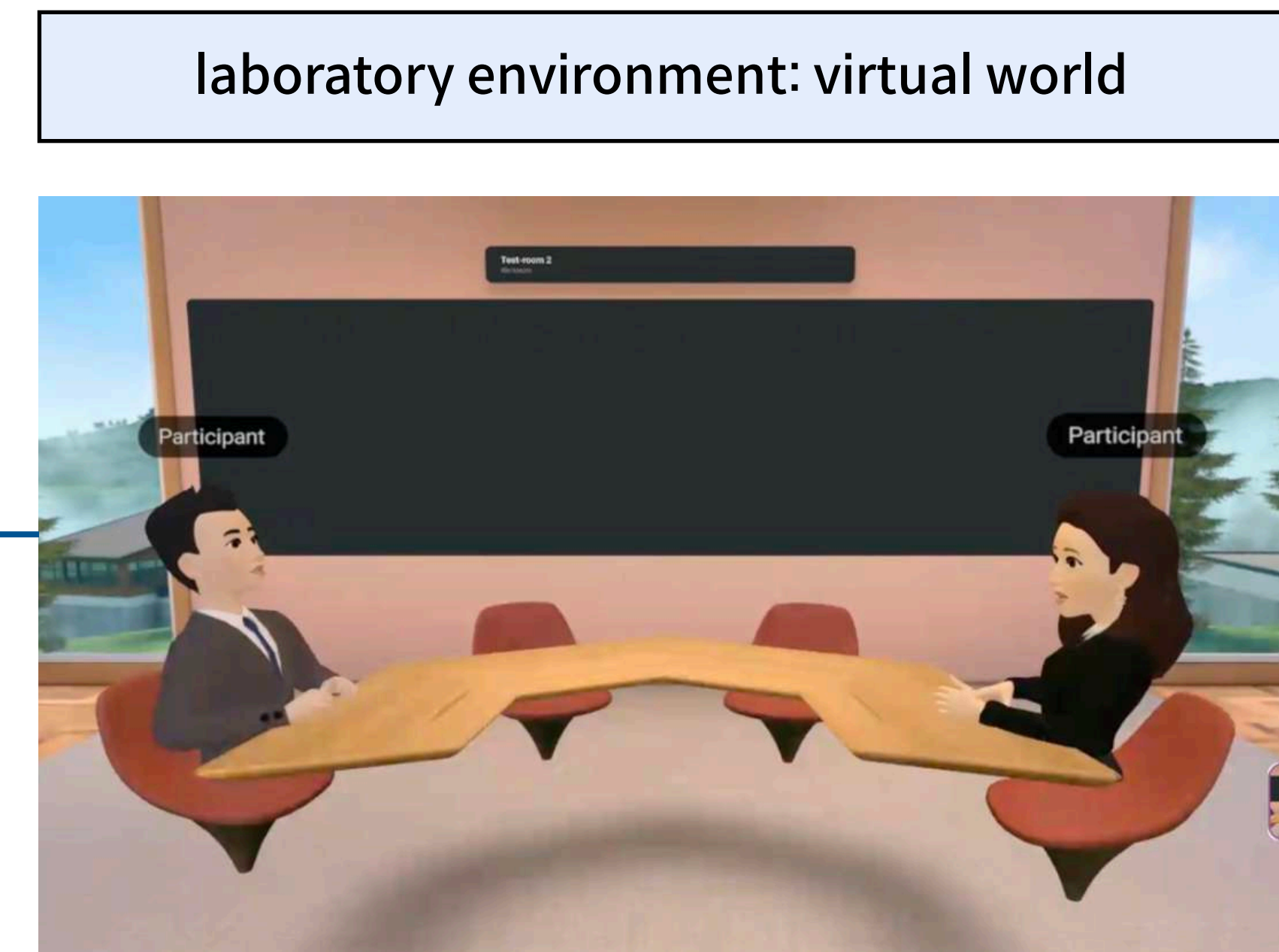
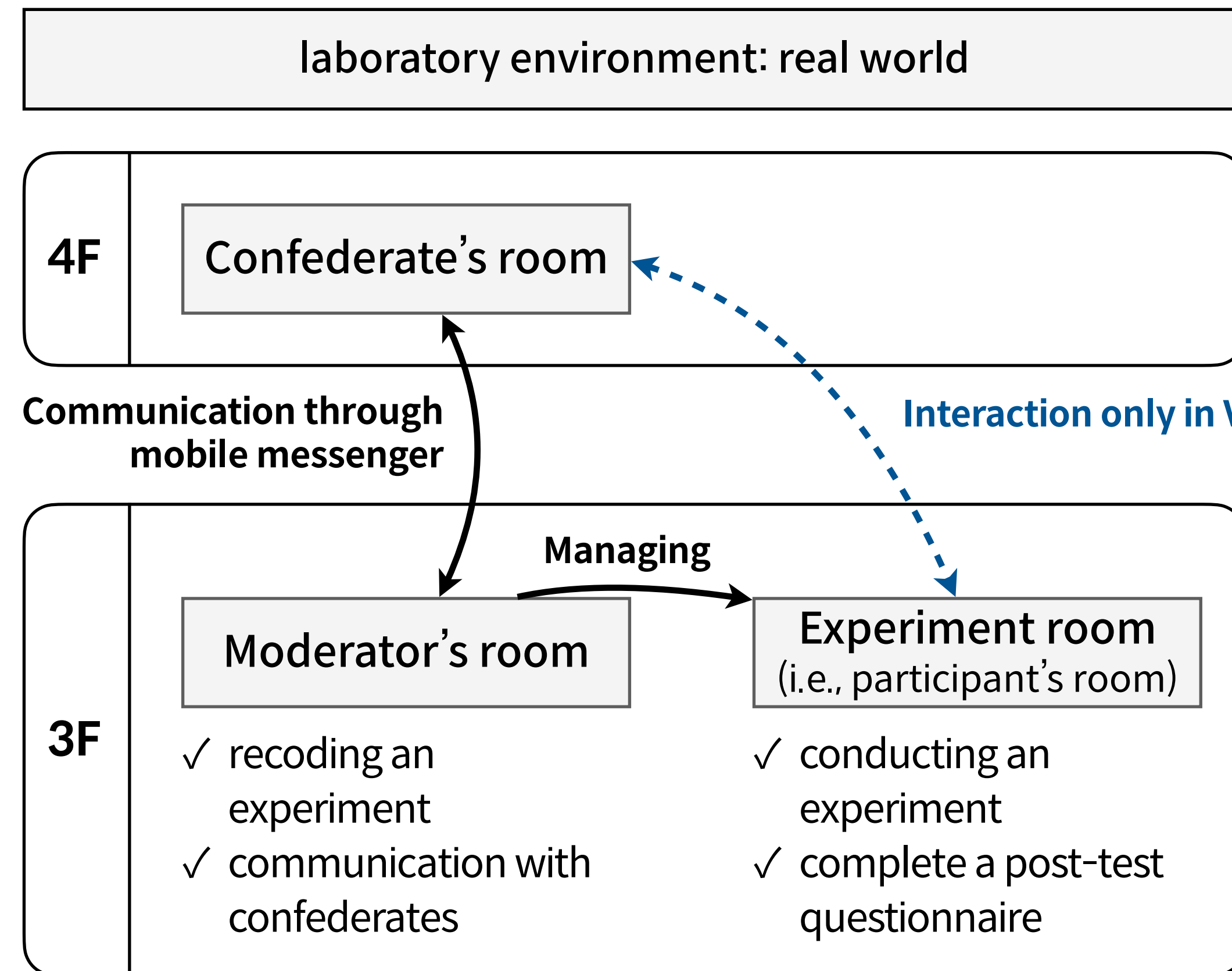
3. Methods

(2) The experimental design and procedure

- The experimental design used a 2x2 design, with the factors being inter-avatar distance(near vs. far) and user's gender(male-female vs. female-male).
- The experimental environment was presented using Horizon Workroom, a commercial VR platform operated by Meta, via an Oculus Quest 2. (Participants viewed the virtual environment from a first-person perspective of their avatar.)
- Each trial involved three individuals: a moderator, a confederate, and a participant.
 - The moderator managed the participants and presented the task scenario in the IVE.
 - The confederate pretended to be a participant and communicated with the participant.
- Participants were given no indication that their interaction partner (i.e., confederate) was in the same building. They were told that they would have to communicate with another undergraduate at a remote location

3. Methods

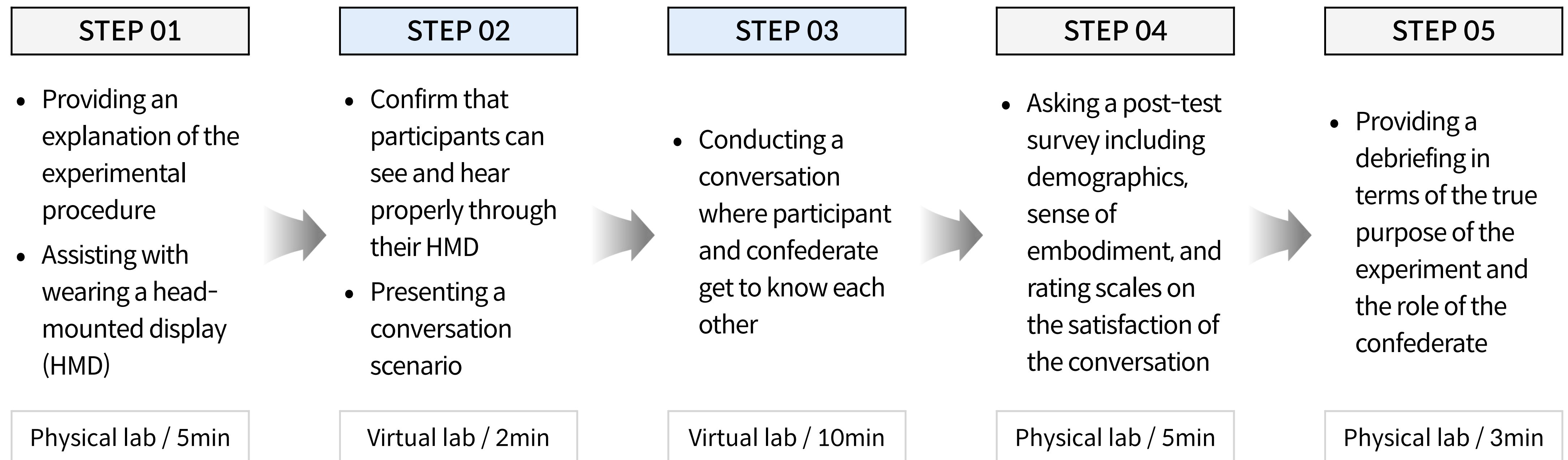
(2) The experimental design and procedure



3. Methods

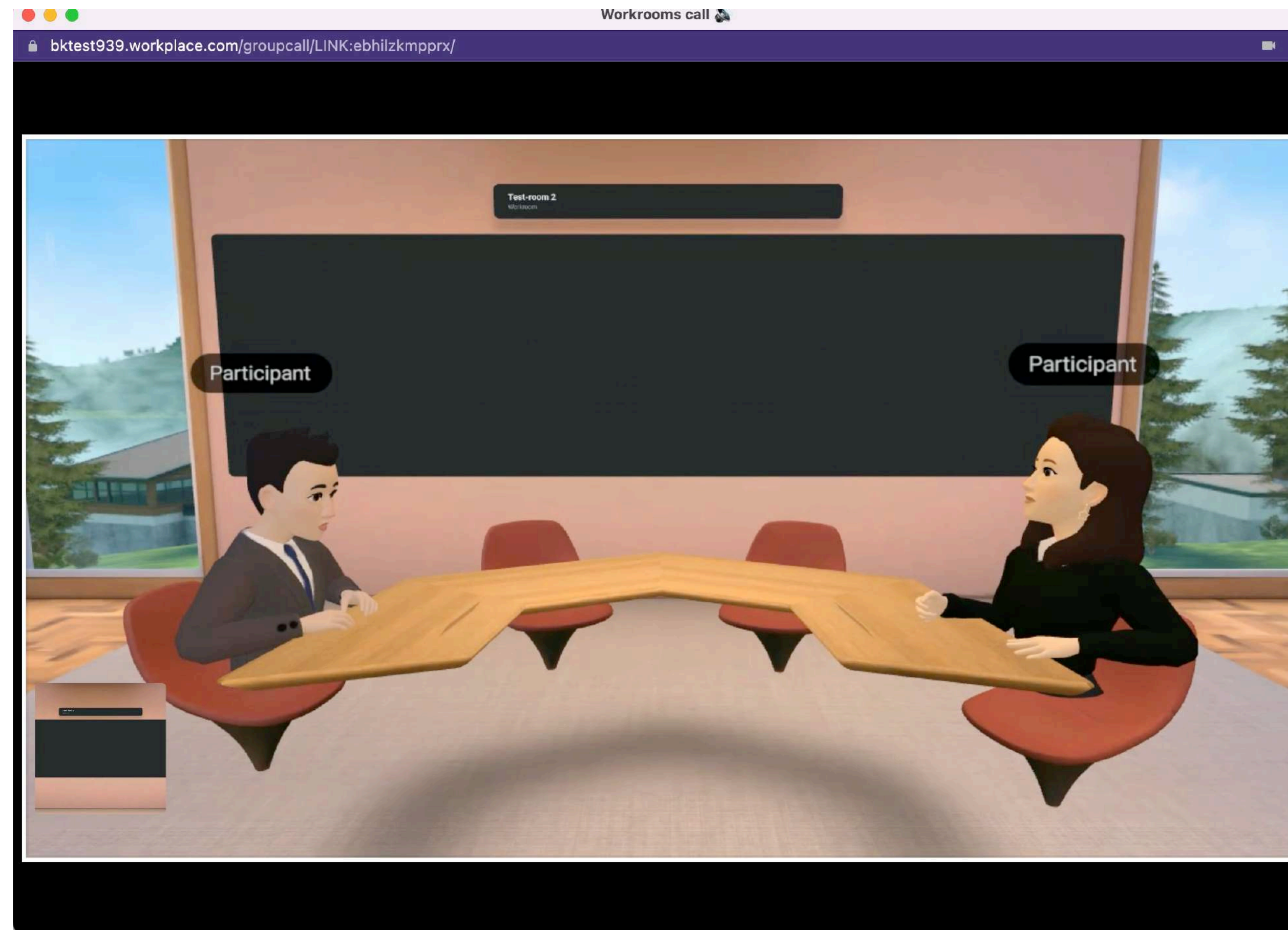
(2) The experimental design and procedure

- Drawing on previous research on interpersonal interaction, a get-acquainted exercise (Frank & Gilovich, 1989) was used for the experiment. The experimental procedure comprised the following steps:



3. Methods

(2) The experimental design and procedure



3. Methods

(3) The experimental manipulation

- The experimental manipulation of inter-avatar distance was operationally defined as shown in figure.
- In the near distance condition, the two avatars were manipulated to be sitting in the middle of the table, next to each other.
- In the far distance condition, the two avatars were manipulated to be sitting at opposite ends of the table, two seats away from each other.



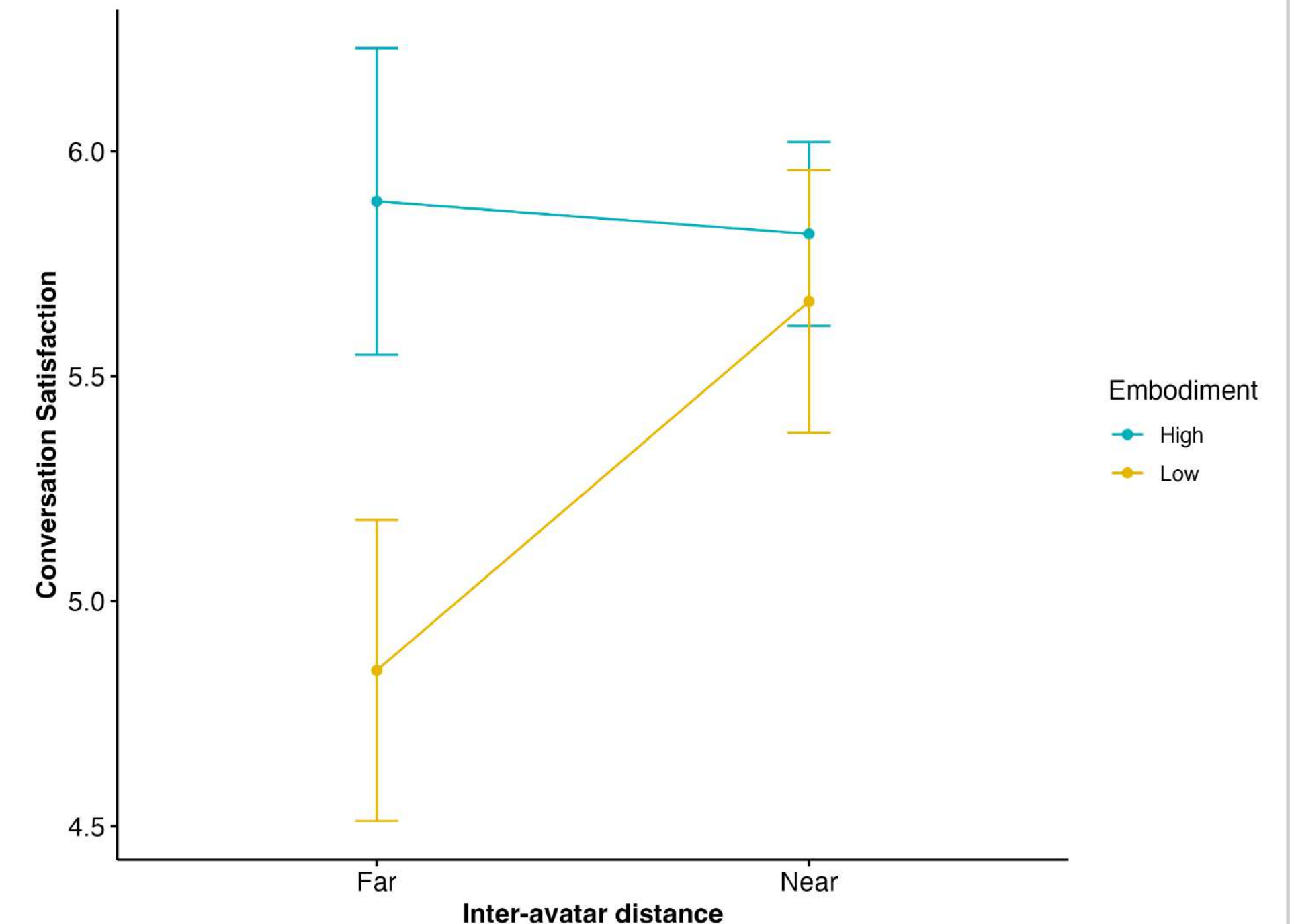
4. Results

(1) The two-way between subject ANCOVA: IAD and embodiment[†]

- A two-way ANCOVA was performed to examine the effects of IAD and embodiment on conversation satisfaction after controlling for avatar's attractiveness.

Source	SS	df	<i>F</i>	<i>p</i>	ω^2
EMB	5.312	1	4.520	.038*	0.049
IAD	2.185	1	1.859	0.178	0.012
EMB*IAD	3.859	1	3.284	0.075	0.032
ATT	5.254	1	4.472	.039*	0.049
Residuals	65.803	56			0.05

Note. SS = sum of squares; EMB = sense of embodiment; IAD = inter-avatar distance; ATT = avatar's attractiveness; * $p < .05$; Basic rules of thumb are: small effect: $\omega^2 = 0.01$; medium effect: $\omega^2 = 0.06$; large effect: $\omega^2 = 0.14$.

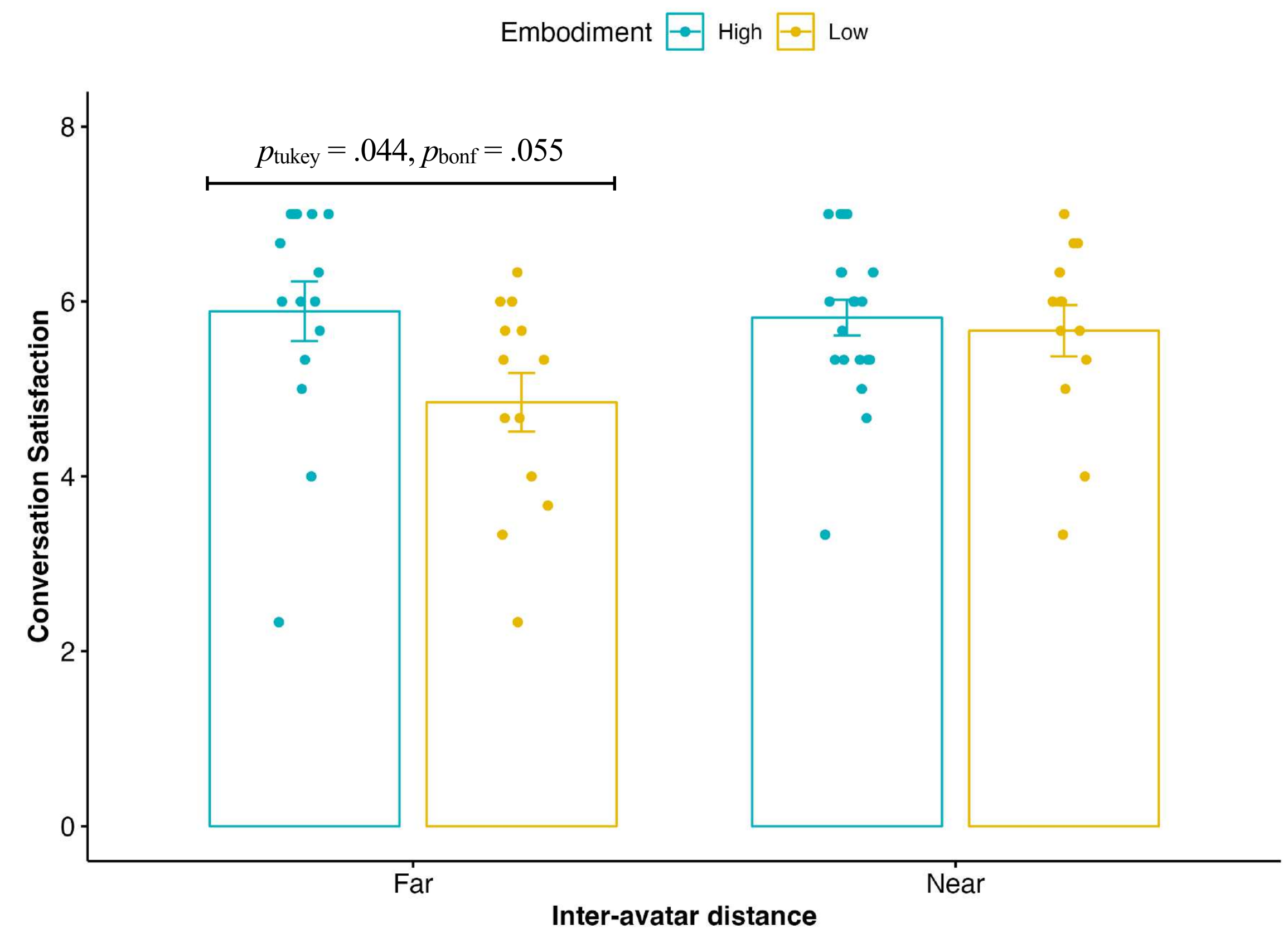


[†] Before conducting ANCOVA, the normality and homoscedasticity assumptions were tested, and there was no significant violation.

4. Results

(1) The two-way between subject ANCOVA: IAD and embodiment

- The simple main effect of IAD was statistically significant in the low embodiment group ($p < .05$), but not in the high embodiment group.
- A Post-Hoc test was performed with Tukey and Bonferroni adjusted.



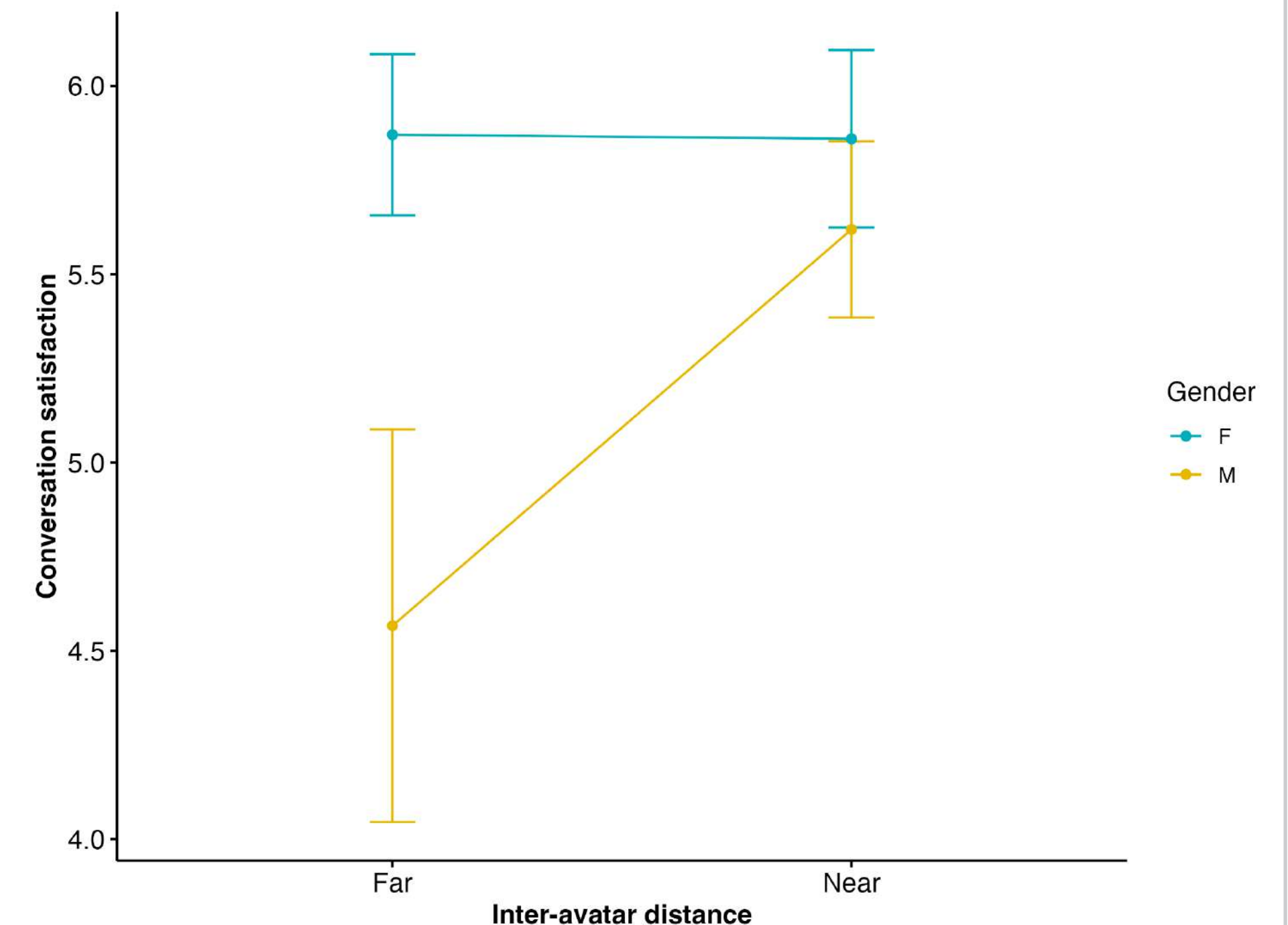
4. Results

(2) The two-way between subject ANCOVA: IAD and gender[†]

- A two-way ANCOVA was performed to examine the effects of IAD and gender on conversation satisfaction after controlling for avatar's attractiveness.

Source	SS	df	<i>F</i>	<i>p</i>	ω^2
IAD	4.295	1	3.893	0.053	0.037
GEN	7.929	1	7.187	.010*	0.079
GEN*IAD	6.011	1	5.449	.023*	0.057
ATT	5.636	1	5.109	.028*	0.052
Residuals	61.777	56			0.05

Note. SS = sum of squares; GEN = gender; IAD = inter-avatar distance; ATT = avatar's attractiveness; * $p < .05$; Basic rules of thumb are: small effect: $\omega^2 = 0.01$; medium effect: $\omega^2 = 0.06$; large effect: $\omega^2 = 0.14$.

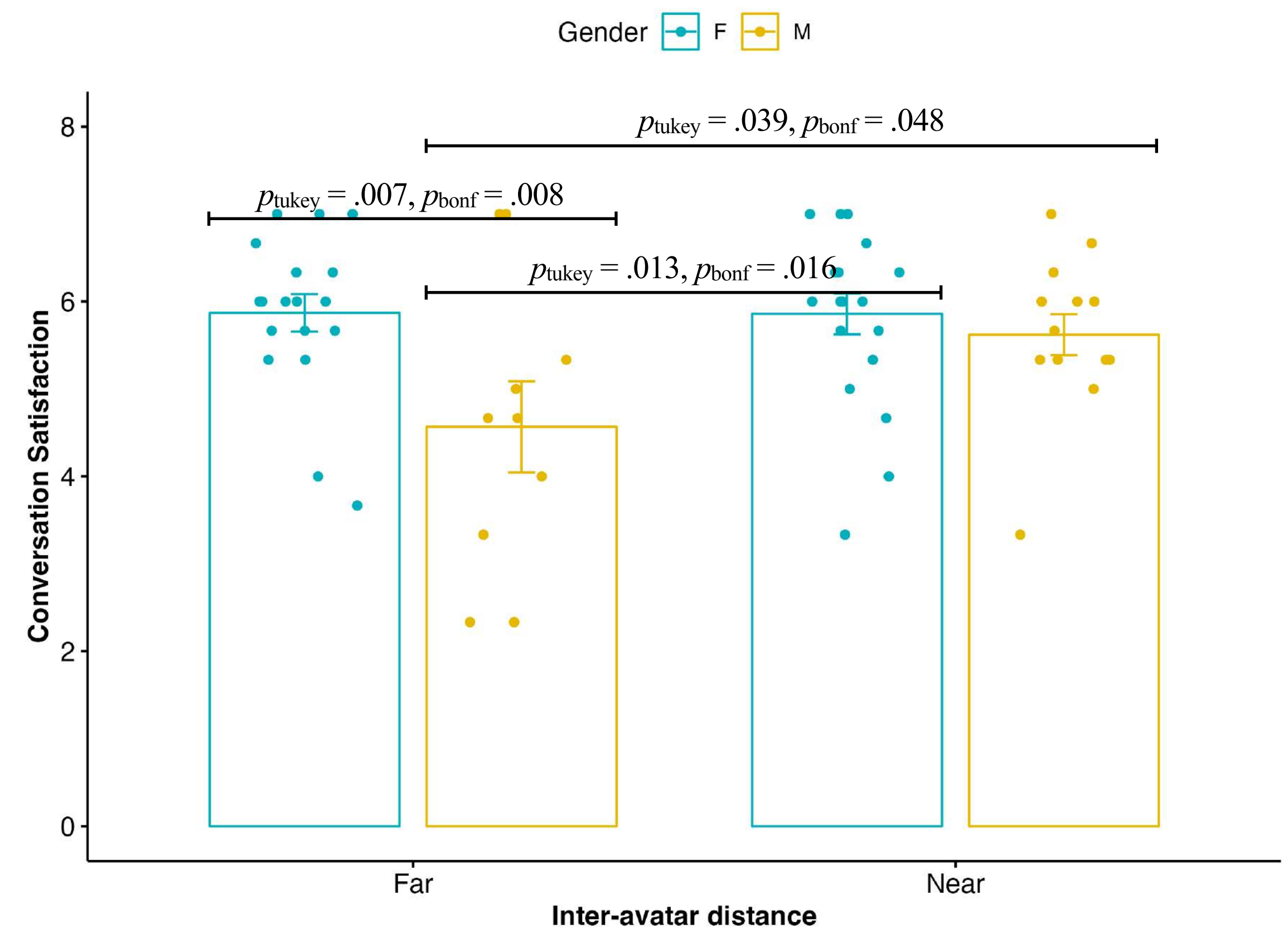


[†] Before conducting ANCOVA, the normality and homoscedasticity assumptions were tested, and there was no significant violation.

4. Results

(2) The two-way between subject ANCOVA: IAD and gender

- The simple main effect of IAD was statistically significant in the male group ($p < .01$), but not in the female group.
- A Post-Hoc test was performed with Tukey and Bonferroni adjusted.



5. Discussions

Implications for research

- This study contributes to the growing body of literature on interpersonal communication in IVE by exploring the role of inter-avatar distance in facilitating effective communication.
- This study used commercial social VR device which may allow the subjects to feel as little as possible like they were in the experimental environment, while producing results that are closest to actual usage.
- Although this experiment has various imperfections, it could fill the gaps in the field of VR communication and has great significance as a pilot experiment.
- Meanwhile, this study provides a very clear relationship of inter-avatar distance and sense of embodiment. It also raises a question that at a strong sensation of embodiment, the effect of distance is not significant.
- Finally, the current study made conversation satisfaction the dependent variable, which is believed to bring some new directions for subsequent studies.

5. Discussions

Implications for practice

- This study's outcomes show that inter-avatar distance truly exists in the IVE. Thus, it must be stricter to prevent the possibility of violating other people's personal space when designing social VR platforms.
- Also, if we want users to have a better experience, we need to enhance their embodiment through all the details.
- These findings may have practical implications for the design and development of VR technologies, particularly those intended for social interaction.

5. Discussions

Limitations

- Due to methodological difficulties in conducting experiments, the study's sample size was relatively small, resulting in overall poor statistical power.
- The demographic limitations of the participants may potentially restrict the generalizability of the findings. Future studies could expand the age range of subjects or conduct experiments with more frequent VR users.
- This study did not perform a manipulation check on distance, so it is possible that the perceived distance of the subjects differed from the experimental manipulation.
- The present investigation solely evaluated VR communication using conversation satisfaction as the dependent variable. Future studies may produce more precise findings if VR communication is assessed using multiple variables.

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